fair → hilesiztoss → yaz

Bugün 21:05 Kategori yok

fair → hilesiz

toss → yazı-tura atmak

sets → kümeler

subsets → alt kümeler → "$A \subset B$"

$\mathbb{Z}$, integers $\mathbb{N}$, natural numbers

$\mathbb{R}$, real numbers $\mathbb{Q}$, rasyonel $\frac{a}{b}$ $b \neq 0$

Superset → A B'yi kapsıyor demek

Universal set "Ω"

Union - "$\cup$" A or B

Intersection "$\cap$" A and B

Complement $A^c, A', \bar{A}$

Diffrence $A - B$

Disjoint sets $A \cap B = \emptyset$ (empty set)

Partition $A_1, A_2 \dots A_n$ are disjoint

and $A_1 \cup A_2 \dots A_n \equiv \Omega$

De morgan's law $\rightarrow (A \cap B)^c \equiv A^c \cup B^c$

$\hookrightarrow (A \cup B)^c \equiv A^c \cap B^c$

Distributive law $\rightarrow A \cup (B \cap C) \equiv (A \cup B) \cap (A \cup C)$

$\hookrightarrow A \cap (B \cup C) \equiv (A \cap B) \cup (A \cap C)$

Range $f(x)$'in alabildigi tüm degerler

$f(x)=1$ and $f(x)=2$ olamaz

$f(1)=1$ and $f(-1)=1$ olabilir

$f(x)$ tanımsız olamaz

Countable $\rightarrow$ finite or enumarateable

$\mathbb{N}$ $\{1,2,3 \dots\}$ infinite and countable

$\mathbb{Q}$ $\{0.000\dots1, \dots\}$ infinite and countable

Power set $\rightarrow 2^n \rightarrow$ all subsets

Exhaustive set $\rightarrow A \cup B \cup C \dots \cup N = S$ ise

↳ kesişim önemsiz

Outcome → Çıktı

Event → durum ⊂ Sample space

probability axioms

$P(A) \geq 0$

$P(S) = 1$

countable collection

↳ disjoint

↳ $P(A_1) + P(A_2) = P(A_1 \cup A_2)$

6 rules → $P(A) = 1 - P(A^c)$

$P(\emptyset) = 0$

$P(A) \leq 1$

$P(A - B) = P(A) - P(A \cap B)$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

if $A \subset B$ → $P(A) \leq P(B)$

Calculating prob → $P(A) = \frac{|A|}{|S|}$

Conditional prob. $P(A|B) = P(A \cap B)$

$$\overline{\quad P(B) \quad}$$

Chain rule $P(A \cap B \cap C) = P(C) \cdot P(B/C) \cdot P(A/B C)$

independence $\rightarrow P(A/B) = P(A)$

$\hookrightarrow \dfrac{P(A \cap B)}{P(B)} = P(A)$

independence and conditionals are

diffrent

$\rightarrow$ 3 events $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

law of the total probability

B₁ … Bₙ be a partition

$P(A) = \sum_{i=1}^{n} P(A \cap B_i) \rightarrow (A \cap B_1) \cup (A \cap B_n) = P(A)$

★ Bayes rule $P(A/B) = \dfrac{P(B/A) \cdot P(A)}{P(B)}$

ways $\rightarrow$ yol

Multiplication principle $1 \rightarrow n_1$ $2 \rightarrow n_2$

$k \to n_k$

possible outcomes $= n_1 \times n_2 \cdots n_k$

Choose → seçme

Replacement → Torbaya atmalı

Ordering → sıralı → permütasyon

ordered sampling with replacement

→ k → örnek sayısı

n → toplam küme elemanı

Ordered sampling without replacement

→ $\dfrac{n!}{(n-k)!}$

Unordered " " "

→ $\dfrac{n!}{(n-k)!\,k!}$

→ Unordered " with "

$\binom{n+k-1}{k}$ → n → yerleşilebilen yer sayısı

k → yerleşen sayısı

Binomial Coefficient

$$\hookrightarrow (a+b)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot a^k \cdot b^{n-k}$$